

Digital FET, N-Channel

FDV301N, FDV301N-F169

General Description

This N-Channel logic level enhancement mode field effect transistor is produced using onsemi's proprietary, high cell density, DMOS technology. This very high density process is especially tailored to minimize on-state resistance. This device has been designed especially for low voltage applications as a replacement for digital transistors. Since bias resistors are not required, this one N-channel FET can replace several different digital transistors, with different bias resistor values.

Features

- 25 V, 0.22 A Continuous, 0.5 A Peak
 - $R_{DS(on)} = 5 \Omega @ V_{GS} = 2.7 V$
 - $R_{DS(on)} = 4 \Omega @ V_{GS} = 4.5 V$
- Very Low Level Gate Drive Requirements Allowing Direct Operation in 3 V Circuits. $V_{GS(th)} < 1.06 V$
- Replace Multiple NPN Digital Transistors with One DMOS FET
- This Device is Pb-Free and Halide Free

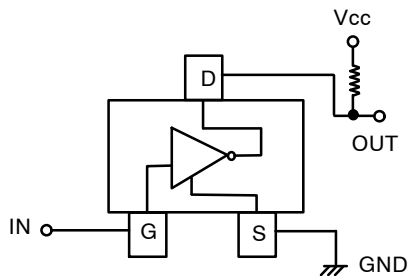
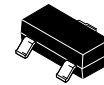
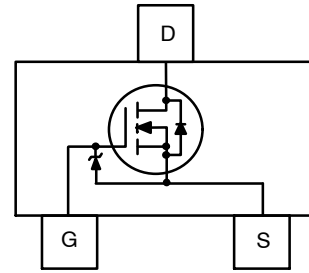
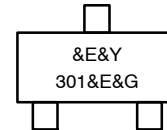


Figure 1. Inverter Application



SOT-23
CASE 318-08

MARKING DIAGRAM



&E = Designates Space
&Y = Binary Calendar Year Coding Scheme
301 = Specific Device Code
&G = Date Code

ORDERING INFORMATION

Device	Package	Shipping†
FDV301N, FDV301N-F169	SOT-23-3 (Pb-Free, Halide-Free)	3000 / Tape & Reel

†For information on tape and reel specifications, including part orientation and tape sizes, please refer to our Tape and Reel Packaging Specification Brochure, BRD8011/D.

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ABSOLUTE MAXIMUM RATINGS $T_A = 25^\circ\text{C}$ unless otherwise noted.

Symbol	Parameter	FDV301N	Unit
V_{DS}, V_{CC}	Drain-Source Voltage, Power Supply Voltage	25	V
V_{GS}, V_I	Gate-Source Voltage, V_{IN}	-0.3 to 8	V
I_D, I_O	Drain/Output Current – Continuous	0.22	A
		0.5	
P_D	Maximum Power Dissipation	0.35	W
T_J, T_{STG}	Operating and Storage Temperature Range	-55 to 150	$^\circ\text{C}$

Stresses exceeding those listed in the Maximum Ratings table may damage the device. If any of these limits are exceeded, device functionality should not be assumed, damage may occur and reliability may be affected.

THERMAL CHARACTERISTICS $T_A = 25^\circ\text{C}$ unless otherwise noted.

Symbol	Parameter	Value	Unit
$R_{\theta JA}$	Thermal Resistance, Junction-to-Ambient	357	$^\circ\text{C/W}$

INVERTER ELECTRICAL CHARACTERISTICS $T_A = 25^\circ\text{C}$ unless otherwise noted.

Symbol	Parameter	Test Conditions	Min	Typ	Max	Unit
$I_{O(off)}$	Zero Input Voltage Output Current	$V_{CC} = 20\text{ V}, V_I = 0\text{ V}$	–	–	1	μA
$V_{I(off)}$	Input Voltage	$V_{CC} = 5\text{ V}, I_O = 10\text{ }\mu\text{A}$	–	–	0.5	V
$V_{I(on)}$		$V_O = 0.3\text{ V}, I_O = 0.005\text{ A}$	1	–	–	
$R_{O(on)}$	Output to Ground Resistance	$V_I = 2.7\text{ V}, I_O = 0.2\text{ A}$	–	4	5	Ω

Product parametric performance is indicated in the Electrical Characteristics for the listed test conditions, unless otherwise noted. Product performance may not be indicated by the Electrical Characteristics if operated under different conditions.

ELECTRICAL CHARACTERISTICS $T_A = 25^\circ\text{C}$ unless otherwise noted.

Symbol	Parameter	Test Conditions	Min	Typ	Max	Unit
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OFF CHARACTERISTICS

BV_{DSS}	Drain-Source Breakdown Voltage	$V_{GS} = 0\text{ V}, I_D = 250\text{ }\mu\text{A}$	25	–	–	V
$\Delta BV_{DSS}/\Delta T_J$	Breakdown Voltage Temp. Coefficient	$I_D = 250\text{ }\mu\text{A}$, Referenced to 25°C	–	25	–	$\text{mV}/^\circ\text{C}$
I_{DSS}	Zero Gate Voltage Drain Current	$V_{DS} = 20\text{ V}, V_{GS} = 0\text{ V}$	–	–	1	μA
		$V_{DS} = 20\text{ V}, V_{GS} = 0\text{ V}, T_J = 55^\circ\text{C}$	–	–	10	
I_{GSS}	Gate – Body Leakage Current	$V_{GS} = 8\text{ V}, V_{DS} = 0\text{ V}$	–	–	100	nA

ON CHARACTERISTICS

$\Delta V_{GS(th)}/\Delta T_J$	Gate Threshold Voltage Temp. Coefficient	$I_D = 250\text{ }\mu\text{A}$, Referenced to 25°C	–	-2.1	–	$\text{mV}/^\circ\text{C}$
$V_{GS(th)}$	Gate Threshold Voltage	$V_{DS} = V_{GS}, I_D = 250\text{ }\mu\text{A}$	0.70	0.85	1.06	V
$R_{DS(on)}$	Static Drain-Source On-Resistance	$V_{GS} = 2.7\text{ V}, I_D = 0.2\text{ A}$	–	3.8	5	Ω
		$V_{GS} = 2.7\text{ V}, I_D = 0.2\text{ A}, T_J = 125^\circ\text{C}$	–	6.3	9	
		$V_{GS} = 4.5\text{ V}, I_D = 0.4\text{ A}$	–	3.1	4	
$I_{D(on)}$	On-State Drain Current	$V_{GS} = 2.7\text{ V}, V_{DS} = 5\text{ V}$	0.2	–	–	A
g_{FS}	Forward Transconductance	$V_{DS} = 5\text{ V}, I_D = 0.4\text{ A}$	–	0.2	–	S

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ELECTRICAL CHARACTERISTICS $T_A = 25^\circ\text{C}$ unless otherwise noted. (continued)

Symbol	Parameter	Test Conditions	Min	Typ	Max	Unit
DYNAMIC CHARACTERISTICS						
C_{iss}	Input Capacitance	$V_{DS} = 10\text{ V}, V_{GS} = 0\text{ V}, f = 1.0\text{ MHz}$	–	9.5	–	pF
C_{oss}	Output Capacitance		–	6	–	
C_{rss}	Reverse Transfer Capacitance		–	1.3	–	

SWITCHING CHARACTERISTICS (Note 1)

$t_{D(on)}$	Turn – On Delay Time	$V_{DD} = 6\text{ V}, I_D = 0.5\text{ A}, V_{GS} = 4.5\text{ V}, R_{GEN} = 50\ \Omega$	–	3.2	8	ns
t_r	Turn – On Rise Time		–	6	15	
$t_{D(off)}$	Turn – Off Delay Time		–	3.5	8	
t_f	Turn – Off Fall Time		–	3.5	8	
Q_g	Total Gate Charge	$V_{DS} = 5\text{ V}, I_D = 0.2\text{ A}, V_{GS} = 4.5\text{ V}$	–	0.49	0.7	nC
Q_{gs}	Gate–Source Charge		–	0.22	–	
Q_{gd}	Gate–Drain Charge		–	0.07	–	

DRAIN-SOURCE DIODE CHARACTERISTICS AND MAXIMUM RATINGS

I _S	Maximum Continuous Drain–Source Diode Forward Current		–	–	0.29	A
V _{SD}	Drain–Source Diode Forward Voltage	V _{GS} = 0 V, I _S = 0.29 A (Note 1)	–	0.8	1.2	V

Product parametric performance is indicated in the Electrical Characteristics for the listed test conditions, unless otherwise noted. Product performance may not be indicated by the Electrical Characteristics if operated under different conditions.

1. Pulse Test: Pulse Width $\leq 300\ \mu\text{s}$, Duty Cycle $\leq 2.0\%$.

TYPICAL CHARACTERISTICS

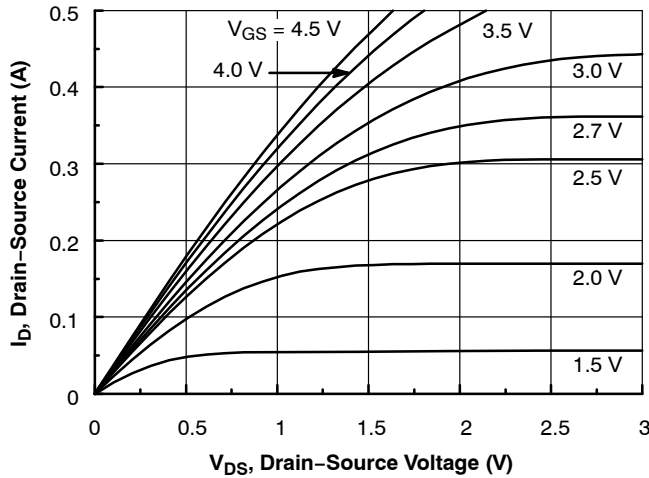


Figure 2. On-Region Characteristics

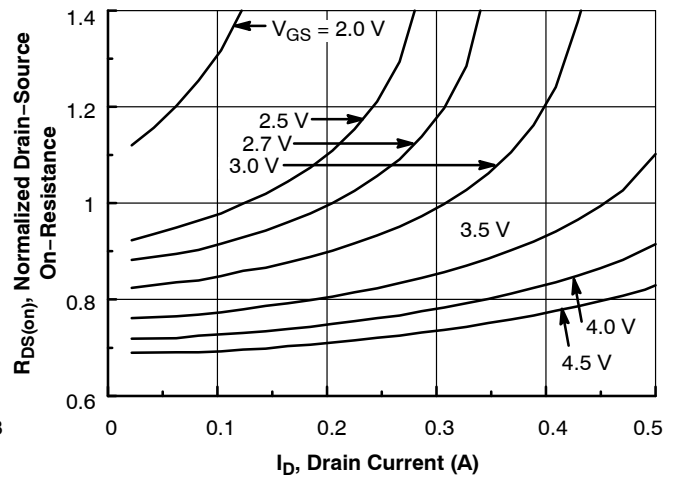


Figure 3. On-Resistance Variation with Drain Current and Gate Voltage

TYPICAL PERFORMANCE CHARACTERISTICS (continued)

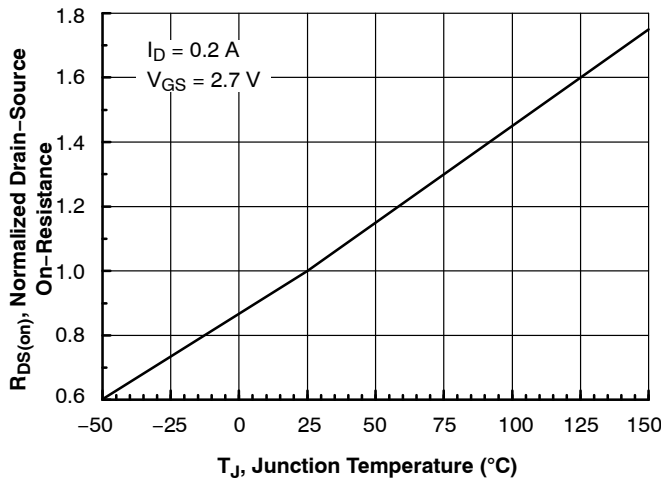


Figure 4. On-Resistance Variation with Temperature

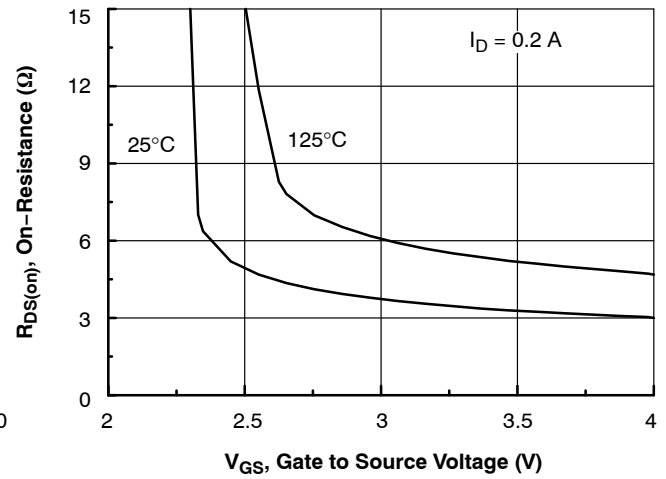


Figure 5. On Resistance Variation with Gate-To-Source Voltage

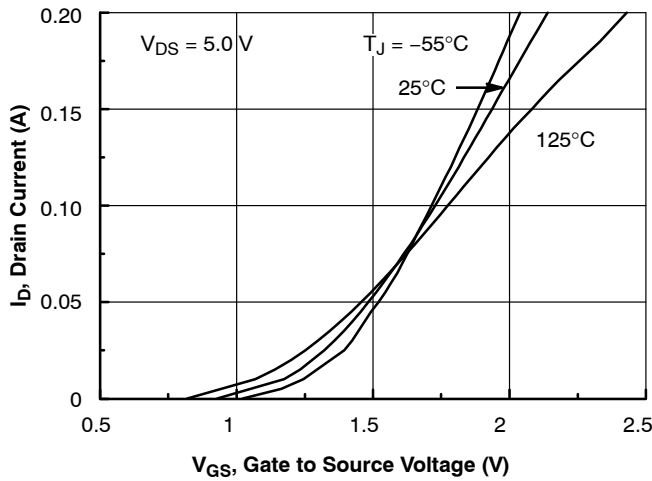


Figure 6. Transfer Characteristics

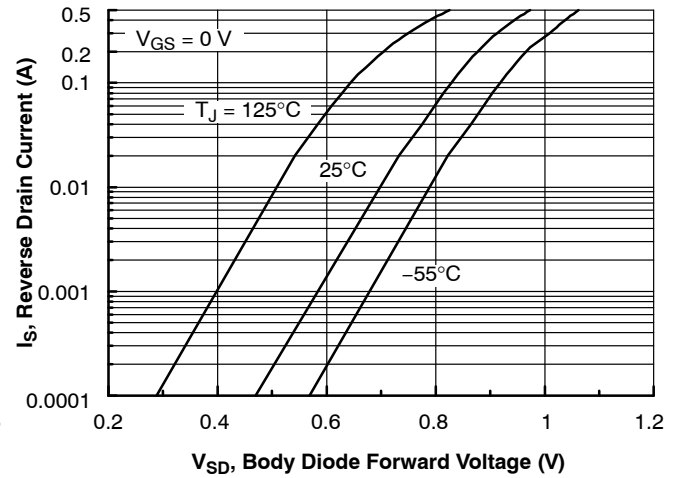


Figure 7. Body Diode Forward Voltage Variation with Source Current and Temperature

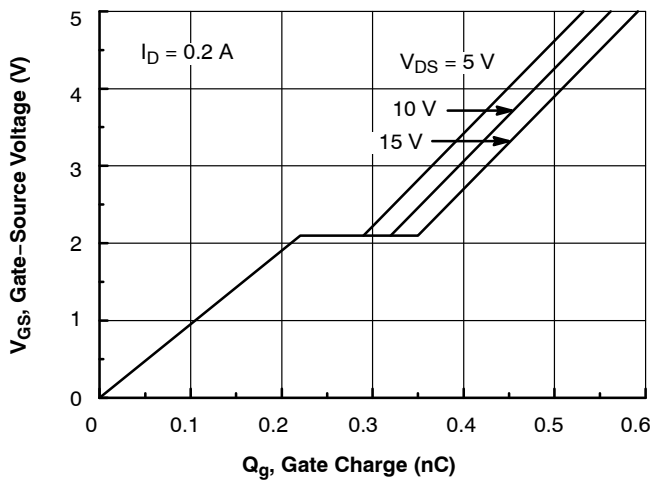


Figure 8. Gate Charge Characteristics

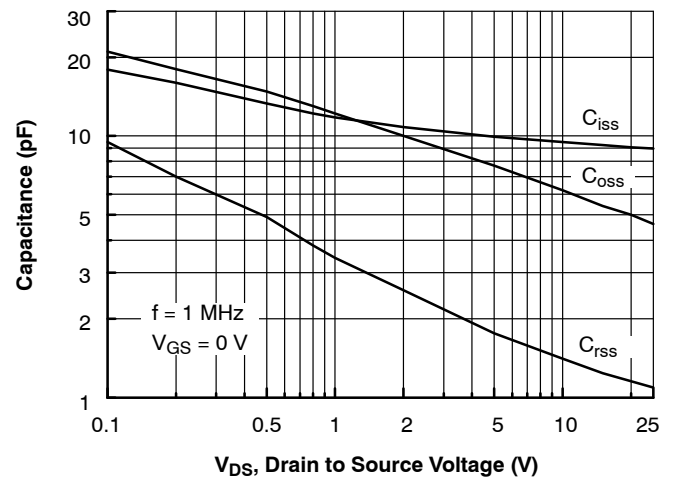


Figure 9. Capacitance Characteristics

TYPICAL PERFORMANCE CHARACTERISTICS (continued)

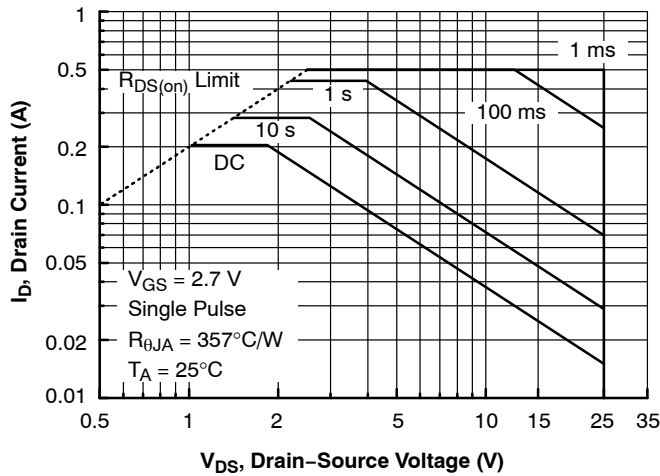


Figure 10. Maximum Safe Operating Area

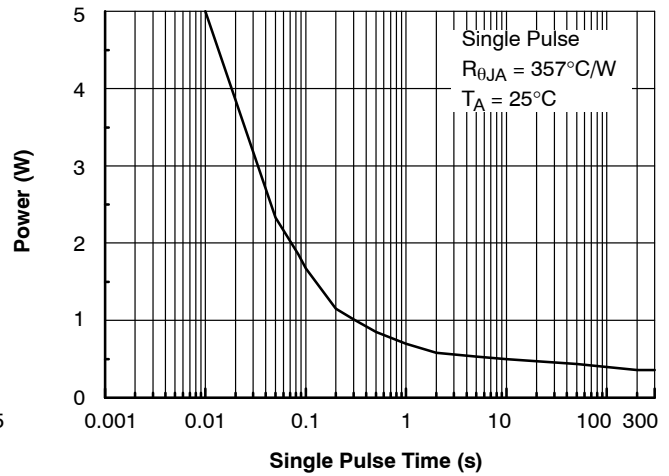


Figure 11. Single Pulse Maximum Power Dissipation

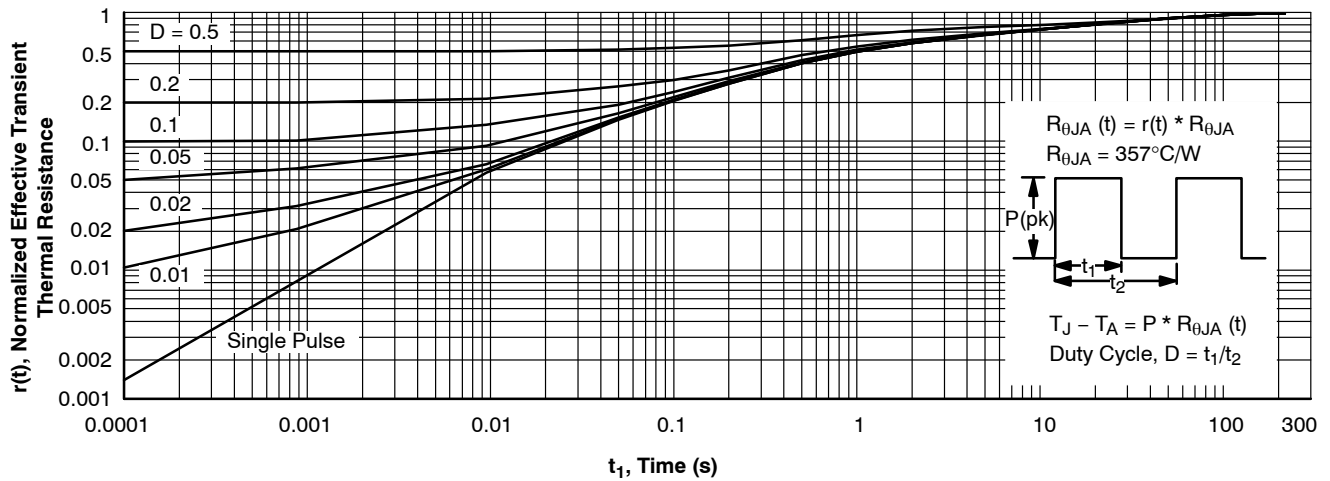
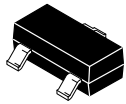


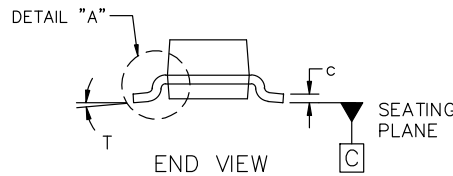
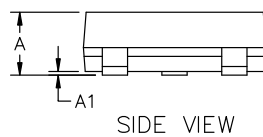
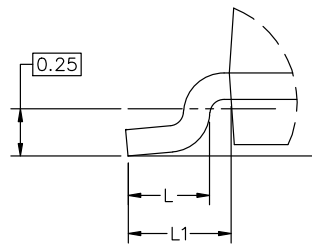
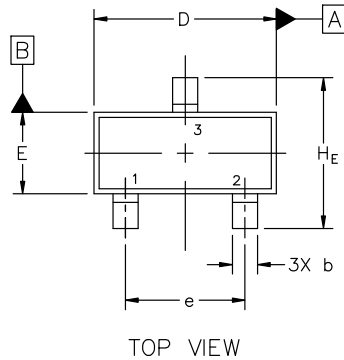
Figure 12. Transient Thermal Response Curve



SCALE 4:1

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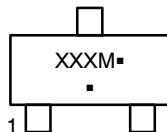


MILLIMETERS			
DIM	MIN	NOM	MAX
A	0.89	1.00	1.11
A1	0.01	0.06	0.10
b	0.37	0.44	0.50
c	0.08	0.14	0.20
D	2.80	2.90	3.04
E	1.20	1.30	1.40
e	1.78	1.90	2.04
L	0.30	0.43	0.55
L1	0.35	0.54	0.69
HE	2.10	2.40	2.64
T	0°	---	10°

NOTES:

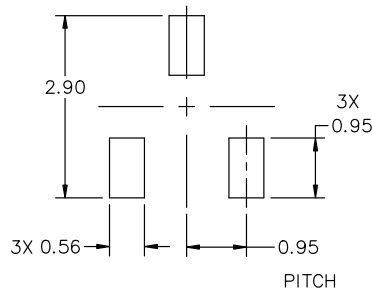
1. DIMENSIONING AND TOLERANCING PER ASME Y14.5M, 2018.
2. CONTROLLING DIMENSIONS: MILLIMETERS.
3. MAXIMUM LEAD THICKNESS INCLUDES LEAD FINISH. MINIMUM LEAD THICKNESS IS THE MINIMUM THICKNESS OF THE BASE MATERIAL.
4. DIMENSIONS D AND E DO NOT INCLUDE MOLD FLASH, PROTRUSIONS, OR GATE BURRS.

GENERIC MARKING DIAGRAM*



XXX = Specific Device Code
M = Date Code
▪ = Pb-Free Package

*This information is generic. Please refer to device data sheet for actual part marking. Pb-Free indicator, "G" or microdot "▪", may or may not be present. Some products may not follow the Generic Marking.



RECOMMENDED MOUNTING FOOTPRINT

* For additional information on our Pb-Free strategy and soldering details, please download the onsemi Soldering and Mounting Techniques Reference Manual, SOLDERRM/D.

STYLES ON PAGE 2

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STYLE 1 THRU 5: CANCELLED	STYLE 6: PIN 1. BASE 2. EMITTER 3. COLLECTOR	STYLE 7: PIN 1. EMITTER 2. BASE 3. COLLECTOR	STYLE 8: PIN 1. ANODE 2. NO CONNECTION 3. CATHODE		
STYLE 9: PIN 1. ANODE 2. ANODE 3. CATHODE	STYLE 10: PIN 1. DRAIN 2. SOURCE 3. GATE	STYLE 11: PIN 1. ANODE 2. CATHODE 3. CATHODE-ANODE	STYLE 12: PIN 1. CATHODE 2. CATHODE 3. ANODE	STYLE 13: PIN 1. SOURCE 2. DRAIN 3. GATE	STYLE 14: PIN 1. CATHODE 2. GATE 3. ANODE
STYLE 15: PIN 1. GATE 2. CATHODE 3. ANODE	STYLE 16: PIN 1. ANODE 2. CATHODE 3. CATHODE	STYLE 17: PIN 1. NO CONNECTION 2. ANODE 3. CATHODE	STYLE 18: PIN 1. NO CONNECTION 2. CATHODE 3. ANODE	STYLE 19: PIN 1. CATHODE 2. ANODE 3. CATHODE-ANODE	STYLE 20: PIN 1. CATHODE 2. ANODE 3. GATE
STYLE 21: PIN 1. GATE 2. SOURCE 3. DRAIN	STYLE 22: PIN 1. RETURN 2. OUTPUT 3. INPUT	STYLE 23: PIN 1. ANODE 2. ANODE 3. CATHODE	STYLE 24: PIN 1. GATE 2. DRAIN 3. SOURCE	STYLE 25: PIN 1. ANODE 2. CATHODE 3. GATE	STYLE 26: PIN 1. CATHODE 2. ANODE 3. NO CONNECTION
STYLE 27: PIN 1. CATHODE 2. CATHODE 3. CATHODE	STYLE 28: PIN 1. ANODE 2. ANODE 3. ANODE				

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